

URVI S SOLANKI

Senior Backend Engineer | AI-Powered Systems | Technical Lead

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Open to Relocation | Open to Remote (EU / US / AU overlap hours) | [Github](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Senior Backend Engineer & Technical Lead with 5+ years building scalable, event-driven backend systems, payment platforms, and AI-powered search and automation pipelines, primarily in Python (Django, FastAPI, Flask). Deep hands-on experience integrating OpenAI, Deepgram, and Pinecone for semantic search, transcription, and automation workflows, alongside payment orchestration across 5+ providers, an 80%+ reduction in PostgreSQL query latency, and Redis caching that cut API response times by 50%. Owns systems end-to-end from tech stack selection and database design to task allocation and client-facing technical decisions while collaborating remotely with US and UK-based clients across overlapping time zones. Twice recognised Employee of the Quarter and once Employee of the Year for high-impact backend delivery.

TECHNICAL SKILLS

Core Stack: Python, Django, FastAPI, Flask, Node.js, TypeScript

AI & Search: OpenAI, Deepgram, Pinecone (vector / semantic search), embeddings, RAG pipeline design

Databases & Caching: PostgreSQL, Redis, MySQL, MongoDB, query optimisation, zero-downtime migrations

Infrastructure & DevOps: Apache Kafka, AWS (S3, SNS, SQS, SES, Lambda), Docker, GitHub Actions, WebSockets, REST API design

System Design: Distributed systems, microservices, event-driven architecture, pub/sub patterns, payment integration, API performance optimisation

Testing: pytest, unittest (Python), Jest (Node.js) 85%+ automated test coverage across critical workflows

Additional Stacks: Express, NestJS, PHP / Laravel production experience

PROFESSIONAL EXPERIENCE

Bacancy Technology Senior Software Engineer (Technical Lead)

March 2025 – Present

Ahmedabad | UK-based Precious Metals Trading Platform

- Own end-to-end technical architecture for the platform from tech stack selection and database design to system structure while directly managing client communication, task allocation, and technical decisions for the delivery team, collaborating remotely with the UK client across overlapping hours.
- Built payment orchestration services integrating 5+ payment providers using FastAPI, PostgreSQL, Redis, and Kafka, enabling reliable transaction processing across multiple payment channels.
- Designed payment orchestration as an admin-configurable system the client switches between multiple payment gateways with zero engineering intervention, cutting feature-change lead time by ~70%.
- Redesigned a real-time pricing distribution system with WebSockets and Redis, eliminating polling and delivering consistent real-time price updates across concurrent user sessions.
- Improved platform reliability through proactive maintenance, monitoring, and query/schema optimisation, contributing to a 25–30% improvement in customer retention.
- Executed zero-downtime migration of 10,000+ user, transaction, and inventory records, maintaining full production availability throughout.
- Designed a centralised, event-driven notification platform for inventory, orders, payments, and customer onboarding with real-time WebSocket delivery, read/unread state, and delivery-status tracking.

Citrusbug Technolabs Backend Engineer

September 2021 – March 2025

Ahmedabad | 3.5 Years | SaaS, AI, Analytics, Marketing Automation Global Clients (US & UK)

- Led end-to-end technical design for multiple client platforms selecting tech stack, structuring databases, and defining system architecture while coordinating directly with clients remotely across overlapping time zones.
- Built 3-5 full backend services using Django and FastAPI across SaaS, AI, and analytics platforms, and developed Node.js/TypeScript (Express, NestJS) services for additional client platforms, delivering production-grade APIs for global clients across both stacks.
- Developed real-time voice-processing workflows and AI-powered automation using Python with Deepgram and OpenAI, enabling transcription, summarisation, and automation pipelines.
- Built a pub/sub-based analytics pipeline (AWS SNS/SQS) tracking email and image-click engagement across marketing campaigns, enabling real-time campaign performance insights.

- Reduced PostgreSQL query latency by 80%+ through execution-plan analysis, indexing strategies, and query rewriting one of the highest-impact performance gains across the engineering team.
- Implemented Redis caching that cut API response times by 50% for frequently accessed workloads across multiple client platforms.
- Integrated payment providers including Stripe, Meta [Facebook], WhatsApp Business API, and Twitter/X, supporting secure payment processing and external-platform integrations.
- Achieved 85%+ automated test coverage using pytest and Jest across critical business workflows, establishing testing standards adopted across backend teams.

ARK Infosoft Data Analyst

June 2020 – July 2021

Ahmedabad | 1 Year

- Built Python-based ETL pipelines processing 100K+ records monthly end-to-end ownership of ingestion, transformation, and loading with error handling and data-integrity validation.
- Automated reporting workflows, reducing manual data-processing effort by ~40% across analytics teams.

KEY PROJECTS

Precious Metals Trading Platform

Bacancy Technology | Mar 2025 – Present

Stack: Python · FastAPI · PostgreSQL · Redis · Kafka · WebSockets · AWS

- Selected the technology stack and designed the core database schema for an admin system that lets the client define custom fields, column types, and dropdown configurations without code changes, cutting engineering requests for routine configuration by ~60%.
- Architected the live metals-pricing service backed by Redis caching and WebSocket distribution, supporting real-time, poll-free price updates across concurrent sessions.
- Built the product-search mechanism using Pinecone vector search over product embeddings, matching on meaning rather than exact keywords for more relevant, typo-tolerant results.

Airyis AI Marketing Automation Platform

Citrusbug Technolabs | Feb 2024 – Mar 2025

Stack: Python · FastAPI · Kafka · AWS SNS/SQS/SES · Facebook · WhatsApp · Twitter APIs

- Designed Kafka-based campaign event pipelines processing multi-channel campaigns across email, Facebook, WhatsApp, and Twitter from a single unified dispatch layer, reducing campaign setup time by ~50%.
- Built a pub/sub analytics layer (AWS SNS/SQS) tracking email opens and link/image clicks across campaigns for real-time engagement reporting a self-driven integration requiring significant research into event-driven tracking design.
- Solved organisation email authentication with a custom AWS SES DKIM integration and a secure token-based verification flow.

Multi-Sports Booking System

Citrusbug Technolabs | Dec 2022 – Oct 2023

Stack: Python · FastAPI · Kafka · AWS SES · SQS · Stripe

- Developed backend services for a multi-sport booking platform supporting court bookings, membership management, event registrations, and daily game scheduling for padel, tennis, and other sports.
- Integrated Stripe enabling membership renewals, split payments among players, and secure transaction processing for court and event bookings.

AWARDS & RECOGNITION

- Employee of the Quarter, Bacancy Technology (Q3 2025) for delivering high-impact backend systems and a new payment orchestration platform while maintaining the existing production platform simultaneously.
- Technical Excellence Spot Award, Bacancy Technology (Q2 2025) for diagnosing and resolving critical performance issues on the precious metals trading platform, enabling it to handle production load while the new system was built in parallel.
- Employee of the Quarter, Citrusbug Technolabs (Q4 2023) for independently delivering critical backend systems and executing a zero-downtime database migration on an aggressive timeline without senior oversight.
- Employee of the Year for Excellence, Citrusbug Technolabs (2024) for consistent high-impact delivery and technical excellence across backend and AI-driven projects, with end-to-end ownership of integration work.

EDUCATION

B.E. Computer Engineering | Gujarat Technological University | CGPA 8.4 | 2018–2022

LANGUAGES

English (Fluent) | Hindi (Native) | Gujarati (Native)